7 Million People. No Good Solution.

BTHS ENABLE INNOVATION HACKATHON

–

05/01/2026

The Problem



- Over 7 million Americans live with visual impairment, with millions relying on braille as their primary method of communication
- Existing text input devices are bulky, expensive, and designed for desks, without real life in mind
- Blind individuals have no discreet, affordable, portable way to type independently in public

Sources: National Federation of the Blind (2026), World Health Organization (2026)

OrCam MyEye
($4000-4500)



Pros:
- Accurate text reading
- Hands free
- Daily task assistance

Cons:
- Steep price
- Limited battery life
- Requires glasses

Verdict
- Powerful but overpriced
- Limited typing, ideal for only reading

Perkins-Style Keyboard
($300-$1500)



Pros:
- Fast typing
- No visual required
- Muscle memory

Cons:
- Learning curve
- Little app integration
- Awkward use

Verdict
- Not portable
- Requires long setup
- Best for typing applications

JAWS ScreenReader
($1000+ License)

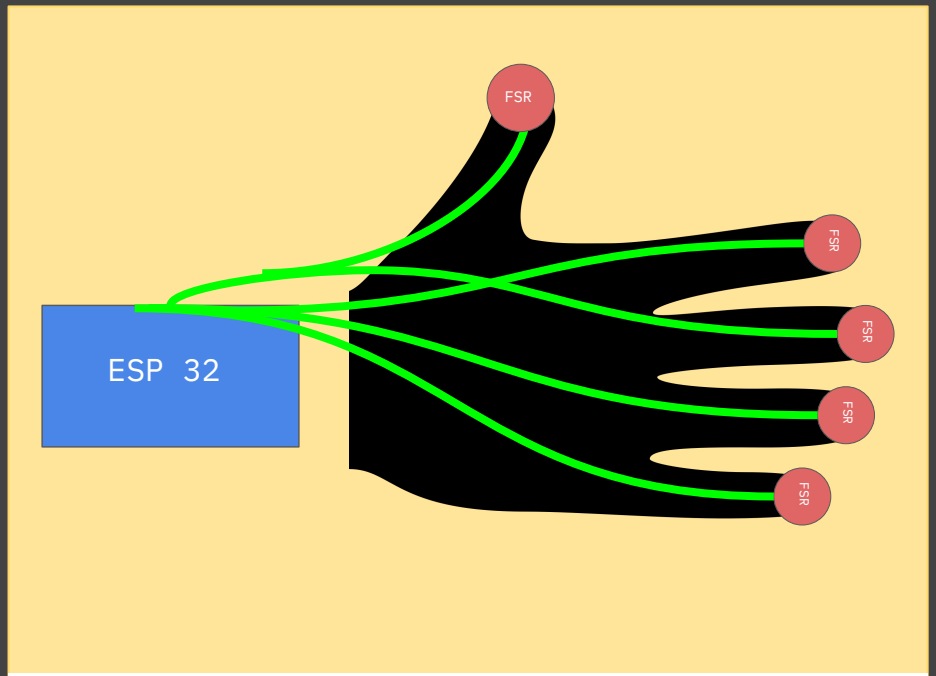


Pros
- Wide support
- Scripting & commands
- Software works with

Cons
- Pricy license
- Windows-only
- Hard to learn
- High system resource use

Verdict
- Great for navigation, not input
- Highly expensive and complex
- Software-only

The Solution



Technical Architecture

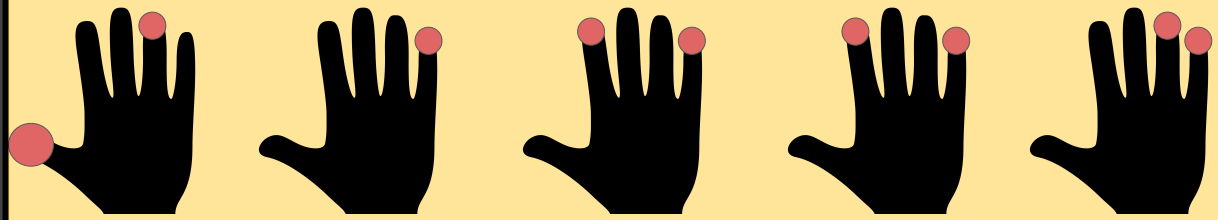
BrailleTap

BRAILLE GLOVE

- ➔ Wearable Braille input glove
- ➔ Fingertip pressure sensors
- ➔ Chorded input to text
- ➔ Wireless output
- ➔ Light and simple design

How it Works

1. Tap finger combination



2. System detects input

H → 0b01001

E → 0b10000

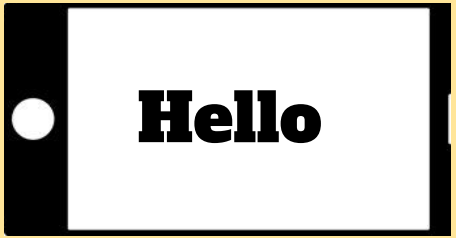
L → 0b10010

L → 0b10010

O → 0b11000

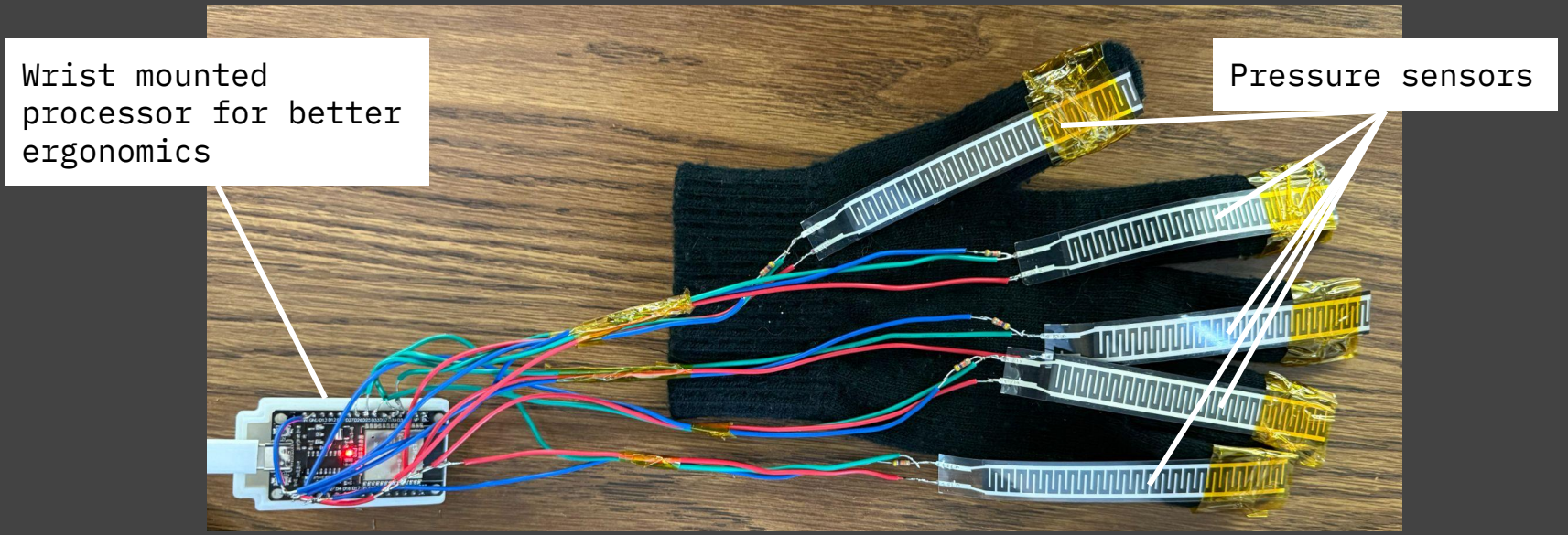
→ ESP 32 →

3. Text sent to display



Prototype Showcase

- [x] Pressure Sensing
- [x] Processing
- [x] Basic Input Mapping



What's Different

Wearability

- Light, durable glove that can fit in your pocket
- Comfortable, non-restricting when not in use
- Completely silent use

Solves – Bulky, noticeable assistive equipment

Affordability

- Targets <$200 vs $1000+ alternatives
- One time purchase not sold with a license
- No cameras or complex mechanics

Solves – High cost of existing tech

Portability/ Ease of Use

- Chorded input = fast learning
- No screen/surface required
- No setup, ready for instant use

Solves – Slow input, reliance on phones/surfaces

Impact

The Problem

- Only 10-25% of blind population are Braille literate
- While only 30% of blind people are employed, 85% of Braille-readers are
- Audio tools, like text to speech, are heavily relied on
- **Limited communication = limits on education and independence**

“If the literacy rate among sighted people were 25%, would that be acceptable?”

– Teacher of the visually impaired

Our Solution

- Faster communication
- Encourages Braille style input
- Makes Braille practical daily
- Affordable + portable access
- **Greater independence**

ACCESSIBILITY → AUTONOMY

Next Steps

Hardware Improvements

- 6th wrist flex input for true Braille
- Tactile Braille output
- Sensor + durability improvements

Design

- Concealed wiring
- Non-glove version
- Smaller, lighter wrist component

Software

- Companion app/customization
- Bluetooth keyboard output
- Input filtering + accuracy
- Braille training modes



Improved gloved iteration



Prospective non-gloved iteration

Typing shouldn't require sight

Touch is enough

Product design - Oscar Tutschulte

Slides & pitch

Positioning & branding

User experience

Product design - William Ferguson Kunstler

Hardware development

Software logic

System integration